THE NUMBER 1 IN FRONT OF THE MIRROR

Mathematical Materiality from the nexus to the cosmos — Ugo Lanciano, October 2026 — v13

THE FRAMEWORK

The question

Physical laws work according to a mechanism that can be expressed in mathematical language: they hold everywhere and always, without exception, and nobody enforces them. An electron here and one in a distant galaxy follow the same rule, exactly. Light that left a galaxy ten billion years ago obeys the same equations as the light of a lamp switched on now. What guarantees them?

The two classical answers leave a gap between the rule and matter. If the law lives in a heaven of forms, as Plato thought, it is unclear how it touches matter. If it lives in our language, as a convenient description we have given ourselves, it is unclear how it guesses the world, and why the world keeps proving it right even where nobody had yet looked.

The simulation hypothesis is the first answer with a programmer in place of heaven: the laws would be the instructions of a program, and someone, somewhere, would be running it. But a program needs a machine, and the machine needs physical laws of its own. Vazza (2025) has estimated that simulating a universe like ours, under its own laws, would require resources that those laws do not allow.

This overview proposes a third answer and follows it, level by level, from the bottom of reality up to the galaxies and the cosmos.

1. The Great Reversal and Mathematical Materiality (MM)

If a rule is always respected and nothing enforces it, then there is not even something underneath: there is only the rule. What we observe, looking at the world, is mathematics itself. A triangle does not obey the rule of having three sides: it is that rule. In the same way, an electron does not obey its laws: it is those laws.

The Great Reversal does not build a bridge over the gap between the rule and matter: it says there is no gap. Matter is not a thing that obeys a relation: it is the relation.

Science has made a move of this kind several times already. Caloric, phlogiston and the vital force seemed substances: they turned out to be names given to processes and relations. Heat is energy passing from one body to another, combustion is a reaction with oxygen, life is organised chemistry. The ether, the medium that was supposed to carry light, disappeared when relativity showed that light does not need one. The cleanest case is the kilogram. Until 2019 it was a cylinder of platinum and iridium kept at Sèvres, near Paris; since then it has been defined through a constant of nature, Planck's constant, that is, through a relation. Each time the world stayed exactly the same: only what we thought lay underneath changed.

Mathematical Materiality (MM), the framework I propose, applies the same move to the last two items left: matter and law.

2. To exist for another

If matter is relation, so is existence. In logic nothing exists on its own: to exist means to exist in relation to another logical entity. A thing exists only for something with which it has something in common, and at the same moment that other thing exists for it. Existence is not a property that a thing possesses in itself, like a colour: it is a mutual relation. A thing fully isolated, with nothing in common with anything else, would not be a hidden thing: it would simply not exist.

For the same reason nothing can exist outside the web of relations of which we are part. Another universe with no link to ours would not exist for us, nor we for it, and one could not even say that it is there. This is why in MM there are no multiverses: there is a single relational structure, and everything that exists is part of it. Of structures outside ours one could not speak: they would lie beyond Wittgenstein's seventh proposition, “whereof one cannot speak, thereof one must be silent”.

This thesis is close to that of Max Tegmark (2008), for whom the universe is a mathematical structure. Two points, however, separate me from him. For Tegmark every mathematical structure exists as a universe; for me only what is in relation exists. For Tegmark the whole collection of structures could be viewed from outside, from a “bird's-eye view”; for me there is no outside. Whoever observes is made of the same mathematics as what they observe, and cannot step out to look at it from outside, as a scientist looks at a test tube: they are inside the test tube.

MM is also close to the structural realism of philosophers of science, in its most radical form: relations are not only what we can know of the world, they are what there is.

In physics the closest position is Carlo Rovelli's relational quantum mechanics (1996): the properties of a system are not absolute but relative to another system, and any physical system can act as an observer. Rovelli starts from systems and from quantum mechanics as already given; I start from the relation itself, and the relation is matter. In chapter 12 the same idea serves to say what the objects we see are.

3. The non-local phase, present everywhere

The usual mistake is to start from an objective space, a ready-made stage on which things stand. I do not start there. I distinguish a non-local phase and a local phase, but they are not two stages, one before and one after, nor two planes laid one on top of the other. The non-local phase has no time, so it cannot come before anything; it has no place, so it does not sit somewhere from which a distance could separate us. It is present in every thing, in every place and at every instant. The local phase does not replace it and does not leave it behind: it is the same reality, read from inside, from a given point.

The first logical nexus is the one between space and time, which with Einstein became homogeneous. In it there is a non-local phase, made only of a logical distinction: space is not time, with no relation of quantitative calculation between the two, except the common logical entity they share, which is needed to make them exist for each other. Space and time, the two distinct terms, share only one logical element, which I call their greatest common divisor (GCD), just as the numbers 3 and 5 share only 1.

The non-local phase is made of pure relations, with no geometry, no space and no time. There are no objects and no distances, only:

- the nexuses: relations between two “logical differents” which exist for each other thanks to what they have in common;
- the two ends of every nexus: the Described, the side that never presents itself as an object, and the Descriptor, the side that counts, measures and opens onto the local;
- the GCD, the common element that makes them exist for each other: it is the only way of counting available to a phase that, having no geometry, cannot count in any other way.

A nexus has the form of the relation between 3 and 5, which have only 1 in common. There is no intermediary between them: nothing travels from one to the other, and the relation holds entirely and at once. And the terms do not come before the relation: the 1 is not a ready-made brick, it appears as the only thing the two have in common. “Non-local”, here, does not mean “at a distance”, nor “faster than light”: it means that space and time are not the container of the relation, but the way it is read.

4. Where it shows: entanglement

If the non-local phase is present everywhere, it must surface in laboratories too, and it surfaces in entanglement. Two systems prepared together, photons or electrons, are measured far from each other: in 2015, in Delft, they were two electrons in laboratories 1.3 kilometres apart, linked by means of photons. The settings of the instruments are chosen at random at the last moment, so that no signal, not even at the speed of light, can carry the choice made on one side to the result on the other.

Bell's theorem (1964) states that, if three assumptions held together, the correlations would have to stay below a certain limit. The three assumptions are: each system carries its own properties with it; no influence travels faster than light; the choices of settings are independent of those properties. Experiments exceed that limit, since 2015 also with the main loopholes closed, and in 2022 Aspect, Clauser and Zeilinger received the Nobel Prize for it. Yet these correlations cannot by themselves send a message. And there is a second, higher limit that no experiment has exceeded: the ceiling predicted by quantum mechanics, the Tsirelson bound.

For me the two systems are not two separate things exchanging signals: they are the two ends of a single nexus, and the nexus is not divided by the distance that separates them in the laboratory. The correlation does not travel: it is the relation itself, read at its two ends.

The ceiling, too, is found in MM. If each end is read locally with the rule of quantum mechanics, the correlations can reach the Tsirelson bound and no further. If instead the two systems carried only a common record written at the start, they would stay below the Bell limit, which experiments exceed (MM 2.48). The ceiling, then, is not set by the non-local phase: it is set by the local reading.

The experiments, by themselves, do not single out MM among the interpretations of quantum mechanics, which predicts them without MM as well. They do show, with data, that the three assumptions of the usual picture cannot all hold together. MM drops the first one at the root: it does not start from separate systems in a space already given, but from a non-local phase present everywhere, and sees in entanglement the case in which that presence becomes measurable.

5. The nexus as a relation between coprime numbers

Five logical postulates describe how a relation without distances presents itself, to whoever reads it, as a world of separate things. They are not proved: they are judged by their consequences, and the consequences are the content of the second part of this overview. I explain them with 3 and 5.

1. The initial difference. At the beginning there is not a thing but a difference: A is different from B.

2. Mutual existence. Two entities exist for each other only through what they have in common; with nothing in common they could not even be called different.

3. Being coprime. Two entities that have been given nothing but being two share no composite parts: they are coprime, like 3 and 5. A larger common part, like the 3 between 6 and 9, would be a third entity and would need one more reason.

4. Unity as the only remainder. Only unity remains in common: the greatest common divisor of 3 and 5 is 1. To belong to both, unity must be counted in each, three times in one and five times in the other. Whatever is repeated in something has parts, positions, distances: it has become local. The local is not a place that already existed: it is what happens to a relation when it has to count itself.

5. No distance between copies. But three copies of unity are the same unity counted three times, and between two copies of the same thing there is no distance. The local carries within itself, disguised as a number, the distance-free bond of which it is made.

So there is no border between the two phases: the local is the non-local counted. Every distance is made of copies between which there is no distance.

6. The matrix

The copies that unity leaves as it counts itself in the two ends form the matrix: the set of repeated copies of one and the same nexus. A single nexus does not yet exist as a world; many copies of it do. The matrix is not something other than the nexus: it is the nexus repeated. With it the local phase appears, the one in which we live and measure; but every copy remains the same nexus, wherever and whenever it is read.

The local world arises when a relation is counted through its GCD: its multiples and its fractions form the matrix. In themselves the copies are all equal: in absolute terms space and time are homogeneous, the same material without any distinction. The distinction arises from whoever reads. For an observer inside it like us, the sapiens, the multiples appear as time and the fractions as space. Counting forward, by multiples, the copies line up one after another, each inheriting the previous one and no longer removable: this is time. Dividing unity into parts, by fractions, the copies stand side by side, all together in what for the reader is the same instant: this is space. “In line” and “side by side” are not properties of the copies: they are the way they are read by whoever is inside the matrix, from a given point.

Einstein's relativity, confirmed by experiment, says this in its own way: for two observers in motion relative to each other the space and time coordinates of the same events mix, and two events simultaneous for one are not simultaneous for the other. The present is the edge of the matrix for

whoever reads it, the point at which for them the line grows longer, and this is why their time has a direction.

The observed objects, while being logically autonomous entities in their local relations, are related to the observer through a logical diffraction, in fractions, of the observer's matrix. In physics the word “diffraction” denotes the way a wave opens out as it passes through a slit; here it denotes the way the observer's matrix opens out into fractions, and in those fractions the object appears. We shall meet it again in the quantum world.

The non-local inside every local thing. This is why the non-local phase is neither a residue nor a distant background: it is recognised in everything that holds the same everywhere. The laws are the same in every place and at every instant because every “here” and every “now” is a copy of the same nexus, and between two copies of the same unity there is no distance. In MM the universality of laws is non-locality seen from inside. Laws need no guardian: 3 and 5 do not “obey” the fact that their GCD is 1, they are that fact. Something that violated the law would not be punished: it simply would not be there.

7. The equals sign as a mirror

In mathematics nothing has meaning on its own. An unknown sign remains mute until we write it equal to something we already know, as in the decipherment of the Rosetta Stone: the hieroglyphs began to speak when they were set beside a Greek text that said the same thing. That equals sign works like a mirror: the sign is reflected in it and finds itself inside what defines it. Only then does it exist. A world that is mathematics does the same with everything it contains.

The same holds for whoever looks. The observer is not a mind: it is any part of the structure that records and compares, and at another time it can be the recorded object. Nor does the observer have anything of its own: it exists because it is reflected in what surrounds and defines it. This is why, when it looks at something else, it does not see it as it is in itself, since there is no “in itself”, but sees its own reflection. The mirror is not an object added to the world: it is the way in which a world that is mathematics makes those inside it exist.

The mirror will return twice in the levels of nature. In the quantum world, when two ways of counting the same object become equal like an image and its reflection (chapter 12). And in dark matter, where the same rule of the mirror, read in the three directions of space, gives two measured numbers (chapters 16 and 17).

8. From counting to measuring

Copies are counted; we, instead, measure in metres, seconds and kilograms. The passage from counting to measurement is what MM calls the translator, and it is there that units of measurement, geometry and constants come in. The speed of light is the simplest example. It is not a barrier but the fixed exchange rate between the two ways in which the reader arranges the copies: how many copies side by side are worth one step forward in the line.

The fraction model. An equality such as 1 = 1 is true but tautological and empty of content. To exist, the 1 must write itself as other than itself, dividing itself into parts of itself. Using a fraction as a structural model:

$$1 = 3/8 + 5/8$$

In this way of writing it:

- the numerator counts what is already there: the occupied places; for the universe, the whole recorded history, from the oldest light we see, the cosmic microwave background, up to today;
- the denominator, the 8, is nothing present, but the grid of possible places: physical emptiness;
- since the numerator is smaller than the denominator, part of the grid remains uncovered, the 5/8: to whoever looks, this portion appears as empty space, the distance that gives things an outline;
- time is the passage from one fraction to the next, where at each step a piece of emptiness becomes real.

The effectiveness of mathematics. The physicist Eugene Wigner spoke of the "unreasonable effectiveness" of mathematics in the natural sciences: why do equations written on a sheet of paper predict the motion of the planets? In the Reversal nothing is unreasonable. The local world is made of counts, and counting is arithmetic: mathematics describes the world because the world is mathematics counted from inside.

So much for the framework. From here on I follow it through the levels of nature. For each I say first what is observed, in the words of known physics; then how MM reads it; then what comes out, with the number and the record in which it can be checked; and finally what remains open. The data come first and MM interprets them. Where MM coincides with known physics, it is compatibility: the same framework recovers what physics has already measured.

THE LEVELS OF NATURE

9. Time and motion

What is observed. Moving clocks run slower. This is measured with cosmic-ray muons, which reach the ground because their time runs slower than ours; with atomic clocks flown on aircraft; and every day with GPS satellites, which without this correction would get positions wrong by kilometres. The factor is Einstein's, and depends only on the ratio between the speed and that of light.

How MM reads it. The time of a clock is the number of copies it counts. An elementary step of the clock has two sides, like the two ends of a nexus: two coprime numbers, which at rest are equal. The clock reads the geometric mean of the two sides, while the laboratory, which adds them, reads the arithmetic mean. With 3 and 5 the clock reads the square root of 15, about 3.87, and the laboratory 4. The ratio of the two means is precisely the slowing of moving clocks predicted by relativity, here for a step at a quarter of the speed of light. When the two sides are equal the two means coincide and the clock is at rest.

The identity is known. It is the k-calculus with which Hermann Bondi explained relativity in 1964, and Kevin Knuth and Newshaw Bahreyni (J. Math. Phys. 55, 112501, 2014) derived the same geometric mean, and from it the Minkowski metric, starting from events already ordered. MM reaches it by another route: the pair is not a measuring device but the two ends of the nexus, and the order of events arises from counting.

What comes out. Records 2.36.1, 2.40 and 2.40.1 test this reading with events that have no place assigned in advance, as in causal set theory (Brightwell and Gregory 1991; Bombelli, Henson and Sorkin 2009). The clearest result is this: copies born at random give Einstein's slowing, while copies arranged on a grid give the arithmetic mean, that is, Galileo's absolute time. In one space dimension, with 200,000 copies, the moving clock follows Einstein to within a few thousandths. In three dimensions, at 0.8 times the speed of light and with about 780 copies, one of the registered tests fell outside the 6% tolerance fixed in advance; the comparison made at equal proper time, instead, is within the errors. The record says so plainly: the criterion registered for the whole series of tests was not met.

What remains open. The structures used are known physics; what MM proposes is the sequence that links them to counting, and that sequence is tested, not proved as a theorem. The test in three dimensions at high speed calls for many more copies than we have used.

10. Mass

What is observed. Mass and time are linked in two ways. The first is Einstein's equivalence of mass and energy: every mass carries a rhythm of its own, the Compton frequency, so regular that in 2013 a group at Berkeley used it as a clock, with caesium atoms (Lan and others, Science 2013). The second is gravity: near a mass clocks run slower, as Pound and Rebka measured in 1959 with light climbing a tower, and as GPS has to correct for every day.

How MM reads it. In the matrix the multiples are read as time. That same multiple, when it is directly contained in the matrix of the point of observation, turns into mass, that is, into concrete matter as it is ordinarily understood. There is therefore a close relation between time and mass, in agreement with Einstein's relativity, which ties mass to spacetime. In simple words: the mass of a thing depends on how many copies are needed, at every instant, to keep it in its place together with everything else.

What comes out. The proper rhythm of every mass is recovered exactly, under one declared condition: that the phase of the count be identified with the action of physics divided by Planck's constant. The algebra that follows is exact; the identification is a postulate (report 2.25, in record 2.27). In the Feynman checkerboard of record 2.40, where a particle advances in a zigzag along the light sides, mass is exactly the rate at which the path turns, and the computation confirms it to machine precision.

What remains open. No mass is derived: the masses of the particles enter as measured data. The link between the multiple in the observer's matrix and the mass we measure in kilograms passes through the translator, and there Planck's constant and the speed of light are taken from known physics.

11. Light and charge

What is observed. Light obeys Maxwell's equations. It has exactly two polarisations and its particle, the photon, has no mass. Electric charge comes in quanta: every charge observed is an integer multiple of that of the electron, while quarks carry two thirds or one third, so that the proton has charge one and the neutron zero. The electron has spin one half: to return to itself one turn is not enough, two are needed.

How MM reads it. A count made on the nodes of the matrix is not enough to give light: report 2.25 showed that the differences of a scalar count alone do not produce a field that propagates like the electromagnetic one. Light lives instead on the links between copies: it is a flux along the sides of the matrix. Charge is a winding, a complete turn that cannot be undone without breaking the relation, and this is why it is an integer.

What comes out. In record 2.27 the flux on the links gives Maxwell-type equations that hold exactly on the lattice, a massless wave and, in three space dimensions, two polarisations. The only external assumption is the metric, taken from known physics. With the same field, record 2.40 recovers the Landau levels and Coulomb's law within a few per cent, and the Casimir effect, the force between two plates in vacuum, at 99.97%. Charge as a winding is an integer and adds up; with three colours and a postulate of fractionality, declared as such, quarks carry +2/3 and −1/3, the neutron is neutral and the difference between proton and neutron is always one (report 2.25). Spin one half, with its two-turn rotation, arises from a Clifford algebra built on the matrix, with the postulates adopted (phase 2.29, in record 2.27).

What remains open. The fine-structure constant, which fixes the strength of the electric force, is free. The metric is imported. The spin algebra does not yet say which spin nature chooses for which particle. And true unification, linking geometry, gravity, matter and electromagnetism, remains to be built: record 2.27 says so in these words.

12. The quantum world

What is observed. A particle sent alone towards two slits arrives on the screen at a single point, but many particles, one after another, draw the fringes of a wave. The probability of each point is given by the square of an amplitude: this is the Born rule. The correlations of entanglement reach the Tsirelson ceiling and go no further (chapter 4). And quantum behaviour is not reserved for the very small: recent experiments see interference with objects of more than 170,000 atomic mass units (Pedalino and others, 2026), while everyday objects always stay in a single place.

Known physics explains with decoherence why large objects stay in a single place. The environment, that is, light, air molecules and heat, continually carries away copies of an object's position, and the fringes are erased before anyone can see them: for a grain of dust spread over a hundredth of a millimetre, about 10^{-31} seconds in air are enough (Schlosshauer 2019). It is the environment that counts, not the apparatus: by heating only some fullerene molecules, which then emit more light, the fringe contrast falls from 47% to 7% and then to zero (Hackermüller and others, 2004). In vacuum and in the cold, on the other hand, even a crystal of 16 micrograms has been put into a quantum state, over very small

distances (Bild and others, 2023). Standard quantum mechanics has no mass at which quantum behaviour stops; collapse models, such as that of Diósi and Penrose, add one, and so far no experiment has encountered it.

Another family of experiments concerns observers themselves. In the “Wigner's friend” scenario one observer, the friend, makes a measurement shut in an isolated laboratory, and a second observer, outside, treats the whole laboratory as a single quantum system. The theorems of Frauchiger and Renner (2018), of Brukner (2018) and of Bong and others (2020) show that some natural assumptions cannot all hold together; among them, that what each observer has recorded holds in the same way for everyone. Experiments with photons, which play the part of the observers, exceed the limits that those assumptions would impose (Proietti and others 2019; Bong and others 2020). When, instead, two observers read the same quantity and compare their records, every experiment has found them in agreement, within the measurement errors.

How MM reads it. The non-local phase and the logical mirror reciprocity between the two ends of the nexus give a reason for the probabilistic behaviour of matter. When the observed object is large enough, its mass fixes it in the matrix of the point of observation: we see it in one place and at one instant. When it falls below a certain mass level, the principal GCD, that of the point of observation, and the GCD of the diffraction that generates the object are forced to become equal, as in a mirror: the object no longer occupies a single point of the spacetime generated by the observer but spreads over several possible positions, and we observe it as probability. It is like having to describe a 10 using only the number 1: the only way is to distribute the 1 over ten positions, one for each unit of the 10. The idea that the passage from quantum to classical behaviour depends on mass is close to that of Roger Penrose and Lajos Diósi; in MM the threshold is the point where the two GCDs, the principal one and that of the local diffraction, become equal.

My most recent premise makes this threshold more precise: the limiting mass, on its own, cannot be determined, because it depends on the fractional distance. At the right distance, even a macroscopic object can show quantum characteristics. In the AI's translation, the fractional distance is the quantity to be distinguished, measured in the cell with which the reader distinguishes positions: about half a wavelength for the light that strikes an object, the diffraction cell of the eye or of the telescope for someone looking at it from afar.

The same framework says what the objects we see are. For whoever observes, an object is not a ready-made thing in a given space, which is then seen: it is a part of their matrix, transposed and projected into fractions. “Transposed” means that it presents itself as something that is elsewhere, at a distance from the observer; “projected”, that only a profile of it is kept, as a shadow keeps the outline of a body and loses the rest. Two observers see the same object because their matrices, while not equal as a whole, are equal in a large percentage, enough to be bound to the same projection. The common object is what the two projections have in common: the trace of one and the same non-local relation, read from two different points, like the two ends of entanglement in chapter 4. What has no non-local link with us casts no shadow in our record, and for us does not exist (chapter 2).

This is also how the two facts of known physics described above are read. In the reading proposed by the AI, decoherence is the moment at which the copies left in the environment are enough to make the

position a common part: the object has been read by the rest of the world, even if nobody is looking at it. And the friend's reading, shut in a laboratory from which no copy comes out, is not absolute (2.48): it becomes common when someone compares it, and on a common part two observers cannot disagree.

What comes out. Three results, in records 2.40 and 2.48.

- The Feynman checkerboard arises from counting with an imaginary weight for each turn, and gives the propagation of the Dirac particle: in one dimension with an error of a few thousandths, in three dimensions with the correct relation between energy, momentum and mass, and in three dimensions the construction works only with spin one half.
- If each end of the nexus is read locally with the Born geometry, the correlations reach the Tsirelson ceiling, 2√2, and go no further; with a common record written at the start they stay below the Bell limit. The most precise data sit at the ceiling (Poh and others 2015: 2.82759 ± 0.00051), and the loophole-free tests lie beyond the Bell limit (Hensen and others 2015; Storz and others 2023).
- The fringes of the two slits are recovered on the checkerboard, in the right position.

For entanglement, record 2.48 also builds a model in which the future contributes to choosing the past, in the form that respects the causality of relativity. It is a path already explored by physics, with the two-state vector formalism of Aharonov, Bergmann and Lebowitz (1964) and Cramer's transactional interpretation (1986).

For the Born rule, record 2.48.1 (26 October 2026) tested the counting of copies, with criteria registered before the computations and checked by a second assistant. In the model of 2.40 the Born rule follows from counting copies of equal modulus, as in Zurek's derivation (2005), with three declared assumptions: that probabilities exist, that copies of equal modulus count the same (in the AI's translation, this is my premise of the distributed descriptor 1) and that the probabilities of distinct copies add up. The square comes from the coin of 2.40, which conserves the sum of the squares and, among the sums tested, only that one. Counting the paths as they come, without phases, does not give the dark fringes: that reading does not follow from the data.

For the common objects, record 2.62 wrote eight mathematical formulations of the idea, registered before the computations, and tested them on the same kind of model as Zurek's quantum Darwinism, in which an object leaves copies of its position in the light it scatters. Two observers who read different photons, with no photon in common, agree except in less than one case in a million: the agreement does not come from shared matter. In the model, where the object has only two values, the common part of the two records behaves like a yes or a no: it is zero as long as few copies have been read, and it is the whole object above a threshold of a few tens of copies (23, with copies that are wrong one time in ten and an allowed disagreement of one in a million). A grain of dust in sunlight leaves its position in the light about a hundred million times in a microsecond: this is why, in everyday life, the existence of objects seems absolute even though it is graded. The closest precedent is Zurek's “relatively objective existence” (2022), for which the existence of a state arises from copies and is a matter of degree.

For the mass threshold, the first part of record 2.66 translated my reading into ten precise criteria: each says which physical quantities the two GCDs are, and therefore where the threshold lies. The translations are the AI's and were registered on 22 October 2026, before the computations. Three do

not follow from the data: objects heavier or more tightly bound than they allow have already shown quantum behaviour, such as the clusters of about 7,500 sodium atoms of 2026 and the 16-microgram crystal of 2023. Five remain possible but cannot be checked today: two close to the Planck mass, one at the mass of the observer and two of the Diósi–Penrose type. The reading with the clock of chapter 9 falls exactly on the record-setting experiment of 2026, but describes a condition of the apparatus, not a threshold of the object. The last, with no threshold of its own, in which the boundary is set by decoherence, is compatible with all the data. A prediction with a deadline of 31 December 2035 is registered for the next experiments.

For the fractional distance, the second part of record 2.66 has given the premise a rule of calculation: eight readings, registered on 25 October 2026 before the computations and checked by a second assistant. In the two readings that hold, the limit is not a mass alone: it is a ratio between the quantity to be distinguished and the cell of the reader's copies, together with the number of copies. The readings of the first part with a threshold close to the Planck mass remain possible, not tested. Two objects of the same mass can stand on opposite sides, like the hot and cold molecules of the 2004 experiment; and among nearby stars Sirius B, heavier, is too small for today's interferometers, while τ Ceti, lighter and farther away, is resolved. The framework thus admits the two senses in which a large object shows quantum characteristics: the state, at a separation small enough compared with the copies of the environment, like the 16-microgram crystal; and the reading, at a distance large enough compared with the observer's cell, like the light of stars that gives fringes in interferometers. The two rules coincide with known physics, decoherence and wave optics. What does not follow is the reading in which, for the state of the object, only the distance from the observer counts: with the same apparatus, changing only the environment changes the interference.

What remains open. The Born rule follows from counting only with the three declared assumptions: that probabilities exist and which outcome occurs are not derived, and the equiprobability of equal copies is, in known physics, precisely the contested step (Schlosshauer and Fine, 2005). The Tsirelson ceiling is found within theorems already published. The definitions of the common object tested in 2.62 use the probabilities of quantum mechanics and, by construction, do not give different statistics; the photon model translates the projection, but the matrix, the nexus and the fractionation have no full counterpart in it yet. Today's experiments do not distinguish MM from standard quantum mechanics. The threshold has no number of its own: in the two distance readings that hold it depends on a ratio and on the number of copies, not on mass alone, while the thresholds close to the Planck mass remain possible and untested (2.66). The model requires a number of copies for a reading to become fixed; the data examined do not count them. To decide between a threshold close to the Planck mass and no threshold of its own, one would need a “Wigner's friend” with a mechanical record of 1.5 to 2.2 milligrams: it is a proposal of 2.66, not yet an experiment.

13. Particles, forces and constants

What is observed. The Standard Model describes three forces, the strong, the weak and the electromagnetic, and three families of particles. It works extremely well, but leaves free about twenty numbers, masses, strengths of the forces, mixing angles, which are taken from experiment. The

strengths of the forces change with energy, and at the Planck scale, where gravity becomes strong, they approach each other without coinciding.

How MM reads it. The inverses of the strengths of the forces are read as counts per cell: that of the weak force in the unit of time, that of the strong force in the unit of space. Their ratio at the Planck scale is an initial datum of the matrix. The masses of the leptons are read as a balance between the same and the different.

What comes out.

- With the Standard Model run up to the Planck scale, the ratio between strong and weak is 1.0768 ± 0.0014 (with a three-loop calculation, 1.0761 ± 0.0013). It is a number of the Standard Model, not a new result: MM reads it as the ratio between a unit of space and one of time, and the reading is consistent with gravity only if space and time weigh the same (record 2.38).
- The Koide relation among the masses of the electron, the muon and the tau, known since the 1980s, is 0.6666645 ± 0.0000051, almost exactly two thirds. MM reads it as a balance between the same and the different in equal measure. Applied to neutrinos with normal ordering, it gives a sum of masses of 0.0593 ± 0.0003 electronvolts; inverted ordering would give 0.1029, excluded by cosmological limits. The tightest limits today, taken with the standard cosmological model, would however exclude the whole normal ordering, that of standard physics too: for this reason they are not yet a test of this reading (2.38).
- Some structures of the Standard Model admit a common reading with declared added assumptions: the charges of a family, the eight colour states, the massless photon, single-handed particles on a lattice. It is not a derivation of the Standard Model, and the weak angle, the masses and the mixings come from the data (record 2.52).
- The census of record 2.54 lists twenty-five known numbers touched by the same geometry of the two mirrors: eighteen compatible and seven to be checked in the future, which are not twenty-five independent tests.

What remains open. No constant is derived. The strength of the strong force enters as a datum: the readings that returned it through the ratio 1.0768 were discarded by construction, because they already contained it. Hypercharge is not fixed. The masses of the particles are imported.

14. Gravity and compact stars

What is observed. General relativity describes gravity as the curvature of spacetime, and every measurement made so far agrees with it. In the Solar System the Cassini probe measured the delay of radio signals passing close to the Sun, and with it the curvature predicted by Einstein, with a precision of a few parts in a hundred thousand (Bertotti, Iess and Tortora, 2003). Neutron stars are the most compact objects that can be observed without a horizon: two solar masses in a sphere a dozen kilometres in radius, which the NICER telescope measures today with an error of about one kilometre. For a perfect-fluid star whose density does not increase outwards, Buchdahl proved a limit in 1959: the compactness, that is, the ratio of mass to radius in the natural units of gravity, cannot exceed 4/9.

How MM reads it. Outside a static spherical body, the Schwarzschild solution written in isotropic coordinates has two numbers: the conformal factor, which says how much distances are stretched, and the one that says how much clocks slow down. An exact and simple relation holds between them: the first, plus the product of the two, makes 2. It is an already known property of the solution. MM recognises it as two shares of a single count: the share of the place and the share of the distant observer, which together always make the same sum. The slowing of clocks is the ratio of their rhythms (record 2.36).

What comes out.

- The count recovers the two shares, with a discrepancy that becomes very small when the readings are many. To go from the count to the source, however, a transfer postulate is needed, chosen among four possible ones because it is the only one that reproduces general relativity exactly: Schwarzschild is not derived, it is recognised (2.36). With a premise on exchange, 2.36.2 reads the slowing as the probability of escaping a random walk, and the expected proportionality is recovered within 4.4%, while the strictest registered thresholds were not reached.
- Inside matter, 2.34 extends the same relation. It is an assumption that Buchdahl had already written down in 1964, as 2.34.1 acknowledges. Read with Einstein's equations, such a star has different stresses in different directions. It yields a maximum compactness of 0.4662 for a star of uniform density, above Buchdahl's 4/9, which holds for stars with equal pressure in all directions. With realistic equations of state for neutron stars the maximum mass rises by about 5% and the radius of a 1.4-solar-mass star by about a tenth of a kilometre, much less than NICER's error, which is about 0.8 kilometres. Today the reading cannot be distinguished from general relativity.
- 2.34.2 proves a rigorous ceiling, 12/25, for all densities that do not increase outwards; 0.4662 is conjectured to be the maximum value. Read as matter, the closure has, in the post-Newtonian limit, the same parameters as general relativity.
- All the predictions registered for the stars have been compared: six out of six for mass and radius, five out of five with causal equations of state, three out of five for the binding energy.

What remains open. There is not yet an action or a dynamics: the rule inside matter is an assumption, not a derivation. The choice of the isotropic variable matters: in other coordinates one would obtain another closure. Einstein's equations, as a whole, are not derived from counting.

15. The Solar System and wide binaries

What is observed. In the Solar System Newton and relativity work perfectly. The Cassini probe has set a very tight limit on a possible extra force due to the field of the Galaxy: $(1.6 \pm 1.8)\cdot 10^{-27}$ per second squared (Park and others, 2026). Pairs of stars very far apart, thousands of astronomical units, feel a mutual acceleration smaller than the scale of galaxies, and there the question is open: with data from the Gaia satellite (DR3) some groups see a strengthening of the attraction of about 40%, others see Newton. The next release, Gaia DR4, is expected on 2 December 2026.

How MM reads it. Before DR4, MM registered two different readings of wide pairs. The first, the “chain” of 2.56, carries the scale of galaxies, $cH_0/6$, down to the pairs with MOND's field equation borrowed, and predicts a strengthening of 33–36%. The second is Newtonian, as in 2.54 and in the cell corrected by

the erratum to 2.23. 2.60 adds a third reading, a local one: it is the body that dominates a relation that reads the external field. The Sun, which dominates the Solar System, makes it Newtonian; in pairs of similar stars nobody dominates, and the strengthening remains.

What comes out.

- Carried down to the Sun, the chain of 2.56 gives an extra force seventeen standard deviations above Cassini's measurement, and a motion of Saturn's perihelion that the ephemerides exclude. It was already known, within MOND's equation, that Cassini forces wide pairs to be Newtonian (Desmond, Hees and Famaey 2024; Park 2026). The outcome therefore affects the reading that carries MOND down to the Sun, not the framework (MM 2.60).
- The local reading holds together Saturn (an extra force of about 10^{-30}), wide pairs (a strengthening of 27%) and the weak gravitational lensing of galaxies, where it agrees with the data much better than Newton. It also makes two distant globular clusters, Palomar 14 and NGC 2419, Newtonian. It is a law of motion for bound relations, not yet a field theory.
- On the DR3 data the dry run of 2.56 favoured the chain by a small margin; 2.60 showed that, repeating the simulations with other random series, the outcome would have been uncertain. For this reason the decision is entrusted to DR4, with rules fixed in advance and repeated over several series.

What remains open. Ultra-faint dwarf galaxies lie 2.5 to 20 times above the prediction of the local reading. DR4 will say which of the registered readings holds: a strengthening, Newton, or neither.

16. Galaxies

What is observed. The stars and gas at the edges of galaxies rotate too fast for the matter that is seen. But there is a surprising regularity: the observed acceleration is fixed by the one produced by the visible matter alone, with a single law for all galaxies (the radial acceleration relation, McGaugh, Lelli and Schombert 2016). Below an acceleration scale of about $1.2 \cdot 10^{-10}$ metres per second squared Newton's law stops working on its own. Milgrom, who in 1983 proposed modified dynamics, MOND, noticed at once that this scale is close to the speed of light times the expansion rate of the universe divided by 2π; Verlinde (2016), with emergent gravity, obtained it as $cH_o/6$. In clusters of galaxies, instead, the law of galaxies alone is not enough.

How MM reads it. Two ends of a reading give the form of that law, which MM adopts from the data as MOND does. The scale is read as a cosmic acceleration, $cH_o/2$, linked to the radius of the visible universe, divided by 3, the square of the mirror's factor $\sqrt{3}$. It is the same rule of the mirror that in chapter 17 gives dark matter.

What comes out.

- The scale $cH_o/6$ is $1.09 \cdot 10^{-10}$ with the expansion rate measured from the cosmic background and $1.18 \cdot 10^{-10}$ with the one measured from nearby supernovae: within the band of the galaxies of the SPARC catalogue, $1.20 \pm 0.02 \pm 0.24$ (2.46). It is a reading chosen knowing the value. The coincidence with Verlinde concerns the coefficient, not the reasoning; today's data do not distinguish $cH_o/6$ from $cH_o/2\pi$, and doing so would require a precision of 1.5%.

- In clusters the scale decreases as the inverse of distance: in twelve clusters out of twelve in the X-COP sample this form does better than a constant scale, with a median slope of −0.97. The test registered in advance had given no indication; the slope was seen afterwards, and the usual mass profiles of clusters also explain it (2.46, 2.46.1).
- If the scale follows cosmic age, it should change with distance. The law registered first, proportional to the inverse of age, was a translation introduced by the AI assistant and not a premise of mine; it is in tension with three sets of data (2.42, 2.42.1). My reading is that the scale follows the expansion rate of the moment: with the MUSE-DARK III data, if the published error bars are one standard deviation, this form does much better than a constant scale, even though it was chosen after seeing those data (2.44).

What remains open. The form of the law of galaxies is taken from the data: each of its empirical successes is also MOND's. The mass deficit of clusters is not resolved. The registered predictions on new clusters and on the scale at distance await data released after the records.

17. Dark matter

What is observed. The cosmic microwave background, the baryon oscillations in the distribution of galaxies, gravitational lensing and clusters all say the same thing: there is about five and a half times more matter that emits no light than there is ordinary matter. The Planck satellite gave 5.364 ± 0.079 for this ratio (2018); the most recent data from the ACT and SPT telescopes, combined with those of DESI, give 5.199 ± 0.040 and 5.254 ± 0.037. No dark matter particle has ever been found.

How MM reads it. Dark matter is a mirror image: a hidden sector, with the same laws as ours, which weighs but is not seen (2.44). Between the two images of the mirror there is a factor √3 for each direction of space. Over the three directions the factor becomes 3√3, that is, 5.196. It is the same rule that, in chapter 16, gives the scale of galaxies: two numbers that standard cosmology treats separately, read with a single rule (2.54).

What comes out.

- 3√3 = 5.196 against 5.199 ± 0.040 (P-ACT-LB2), 5.254 ± 0.037 (CMB-SPA with DESI DR2, at 1.6 standard deviations) and 5.364 ± 0.079 (Planck 2018 alone, at 2.1). Imposed on the Planck and DESI data, the ratio 3√3 costs very little: left free, the data would give 5.22 (2.44).
- Verlinde (2016) obtains the same scale $cH_0/6$ and, for the ratio, a different form, which coincides with 3√3 for certain values of the parameters: today's data do not separate the two (2.54).
- Primordial deuterium, produced in the first minutes, fixes the amount of ordinary matter. With dark matter fixed at 3√3 times as much, the DESI DR2 distances can be read without the cosmic background, at negligible cost; the expansion rate that comes out, 68.52 ± 0.59, is a compatibility, not a new number, and the scale of galaxies comes back as $1.110 \cdot 10^{-10}$ (2.58). The predicted lithium is 3.5 times the observed amount, as in standard cosmology: it is the lithium problem, already known. A mirror at the same temperature as our world, with radiation that weighs, is excluded by primordial helium: the mirror must be colder, or its radiation must not weigh, as 2.44 postulates.

What remains open. 3√3 is not derived: it is an MM reading chosen knowing the data. What the matter of the hidden sector concretely is, is not said. The prediction is registered (chapter 21): if the next data exclude it, what falls is MM's reading of the ratio, not MM.

18. The cosmos

What is observed. The universe is expanding, and since the end of the 1990s we have known that the expansion is accelerating. The standard model, ΛCDM, describes this with a cosmological constant worth about 70% of the energy of the universe. In 2025 the DESI distances, combined with supernovae and the cosmic background, gave a first hint that dark energy may change over time.

How MM reads it. Two readings, deposited at two moments.

- In 2.32 the growth of a network of relational links is translated, with declared postulates, into a dark component. Put to the test of three probes together, the Pantheon+ supernovae, the DESI DR2 distances and the Planck cosmic background, it holds as well as ΛCDM, but with three extra parameters, and the selection criteria still favour ΛCDM. The cosmological parameters remain within about one standard deviation of Planck's.
- In 2.50 the late acceleration is read as a reading of the observer's clock, linked to the growth of structures, in a universe without dark energy: the expansion rate we read is the true one multiplied by a factor that follows the growth of galaxies. The closest precedent is Wiltshire's “timescape” (2007).

What comes out. The reading of 2.50 improves on ΛCDM by 7–12 points of χ^2, almost as much as a dark energy that changes over time, which does a little better; one exponent is fitted to the data. In its consistent version the effective dark energy crosses the value of a cosmological constant at relatively nearby distances, and the growth of structures has σ_8 around 0.80. The history of 2.32 is instead in tension with the DESI DR2 data, by two to three standard deviations; the constraint that prevented it from crossing that value came from the translation of counts into energy and pressure made by the AI assistant, not from my premises.

The time of the matrix and the beginning. In MM the true time is the time of the matrix, in which the initial nexus is present everywhere; what we call the beginning of the universe is a common diffraction, which does not exist objectively in the observer's present mirror. The data of the first minutes, deuterium, helium, the number of light species, are present readings. 2.58 translated this premise into six tests: those that could be checked hold, except the hot mirror of chapter 17, and none solves the lithium problem.

What remains open. The observer's clock is today a phenomenological law, not a derivation: the data do not distinguish it from a substance with the same expansion history, and a specific prediction does not yet exist. The full spectra of the cosmic background remain to be compared. Standard cosmology describes with great precision a universe in which known matter is 5% and the rest has two names, dark matter and dark energy, but not yet a nature. MM proposes a structure in which some of those numbers are readings of the same rule, with their falsifiers registered.

19. A simulation without a machine

For all these reasons the universe behaves like a simulation without being one. It follows exact rules, made of counts, everywhere and at every instant, but no machine runs them and no programmer wrote them: a structure of mathematical relations does not need to be computed in order to exist.

That such a simulation need not be run had already been noted by Tegmark (2008): if the world is a mathematical structure, the computation that describes it does not have to move it forward, because the structure already contains all its instants. Vazza (2025) estimated that running it on a machine would require resources that the laws of the universe do not allow. MM removes the machine and keeps the rules: the initial question, what guarantees the laws, has the same answer as the Great Reversal. Nothing guarantees them from outside, because there is no outside: the laws are the structure.

The programs that accompany the records are tools for checking, not the processor of the world: they are for anyone who wants to redo the computations.

THE BALANCE

20. What holds, what is a reading, what is open

Every result holds under the assumptions declared in its record. In the table the status uses only four words. **Known**: a property or an identity that physics already knew, and that MM recognises. **Derived with assumptions**: it follows from counting, given the postulates or the imported data that the record names. **Reading**: an interpretation chosen knowing the measured value. **Programme**: a step not yet taken.

Level	What comes out	Status	Record
Time and motion	Einstein's slowing from copies born at random; a grid gives Galileo	Known (Bondi, Knuth and Bahreyni); registered tests, one outside tolerance in 3+1	2.36.1, 2.40, 2.40.1
Mass	Proper rhythm of every mass; mass as the rate of turning	Derived with a declared postulate; no mass derived	2.25, 2.40
Light	Maxwell-type equations, massless wave, two polarisations	Derived with the metric imported	2.27
Charge and spin	Integer charge; quarks in thirds; spin one half	Derived with declared postulates	2.25, 2.29, 2.52
Quantum world	Feynman checkerboard; Tsirelson ceiling 2√2; Born from counting copies	Derived; Born with three declared assumptions	2.40, 2.48, 2.48.1
Observed objects	The object as the common part of two records; agreement with different photons	Reading tested on a model; no statistics different from quantum mechanics	2.62

Level	What comes out	Status	Record
Mass threshold	Ten readings of the threshold (three do not follow from the data, five cannot be checked today) and eight of the fractional distance; prediction to 2035	Programme: no number yet	2.66
Constants	Strong/weak ratio 1.0768; Koide; neutrinos 0.0593 eV	Known numbers read; no constant derived	2.38, 2.54
Standard Model	Some structures with a common reading	Reading with added assumptions	2.52
Gravity	$\psi + N\psi = 2$ as two shares of a count	Known; Schwarzschild recognised, not derived	2.36, 2.36.2
Compact stars	Maximum compactness 0.4662; ceiling 12/25	Derived with Buchdahl's 1964 assumption; today indistinguishable from relativity	2.34–2.34.2
Solar System and binaries	Local reading: Saturn Newtonian, wide pairs strengthened	Law of motion registered for Gaia DR4	2.56, 2.60
Galaxies	Scale $cH_0/6$; scale following the expansion rate	Reading; predictions registered	2.42.1, 2.44, 2.46
Dark matter	Ratio $3\sqrt{3} = 5.196$	Reading; prediction registered	2.44, 2.54, 2.58
Cosmos	Expansion read almost as ΛCDM; the observer's clock	Phenomenological reading; the criteria still prefer ΛCDM	2.32, 2.50
Three-dimensional space	The number 3	Programme: still enters as a datum	not deposited

The table also says what is most MM's own. It is not a single number: it is the same pair of ideas, the nexus that exists in its GCD and the matrix of its copies read from inside, returning at every level, from the clock to dark matter. Precisely because the same ideas return at every level, each row says what it depends on: a hidden step would be multiplied across all the levels.

21. What MM does not claim, and the dated tests

What it does not claim. MM does not claim to derive the constants of nature: the fine-structure constant, the strength of the strong force, the masses of the particles, the metric of spacetime and the number of its dimensions enter as data. It does not claim to derive the Schwarzschild solution or the Standard Model, nor the Born rule without assumptions: the Born rule follows from counting copies only with the three assumptions declared in chapter 12. It does not claim to solve the measurement problem of quantum mechanics. It does not claim to do better than the standard cosmological model. It does not claim to have proved its postulates: it proposes them, and asks that they be judged by their

consequences. Volume 2.30 puts it this way: it offers a relational proposal, explicit formalisations and a register of the steps not derived; not a closed theory, nor an already established novelty, nor an empirical superiority over existing theories.

The dated tests. Some readings were registered before the data that will test them came out. For each, what would make it fall is written in advance.

- **Gaia DR4, expected on 2 December 2026.** Wide pairs of stars will decide between the chain of 2.56, the Newtonian reading and the local reading of 2.60, with rules fixed and repeated over several series of simulations. For the local reading four sharp predictions are also registered: no difference linked to the direction of the Galaxy's field, distant globular clusters Newtonian, comets Newtonian, no effect of the external field on gravitational lensing.
- **The next DESI distances and the Simons Observatory.** The prediction on the ratio 3√3 has been registered since 16 October 2026 (2.54): if the new value moves too far away, the reading of the ratio falls. A second test uses DESI with deuterium and without the cosmic background, until 31 December 2028 (2.58).
- **The scale of galaxies measured better.** Until 31 December 2030, the first measurement of the scale of galaxies with an error of $0.06 \cdot 10^{-10}$ metres per second squared or less (2.58); and new data on the scale at distance (2.44) and on at least twenty clusters (2.46).
- **The mass threshold.** A prediction registered on 22 October 2026, with a deadline of 31 December 2035, for the next interference experiments with ever larger objects (2.66). The rules are written in a program that anyone can apply to the papers published up to that date.

If one of these readings falls, that reading falls, not the framework: the bridge between the framework and that datum is rewritten. But a framework that reads many levels of nature with the same few ideas is worth as much as the number of its readings that hold, and this is why the readings are written before the data.

22. Where to read

All the records are on Zenodo. The DOIs below lead to the version cited, and from each record's page one can reach its latest version; the DOI of this overview is the general one, which always leads to the latest version.

- Entry page, with the list of all records: MM 2.27, DOI 10.5281/zenodo.22667516.
- The volume, foundations, bridges and programme of verification: MM 2.30, DOI 10.5281/zenodo.22730712.
- Cosmology: MM 2.32, DOI 10.5281/zenodo.22839525; reproduction kit, DOI 10.5281/zenodo.22840735.
- The corrected relational cell: erratum to 2.23, DOI 10.5281/zenodo.23058946.
- Compact stars: MM 2.34, 2.34.1 and 2.34.2 (DOI 10.5281/zenodo.23055577, 10.5281/zenodo.23569906, 10.5281/zenodo.23595759).
- ψ + Nψ = 2 and the moving clock: MM 2.36, 2.36.1 and 2.36.2 (DOI 10.5281/zenodo.23056025, 10.5281/zenodo.23278965, 10.5281/zenodo.23287965).

- The constants: MM 2.38, DOI 10.5281/zenodo.23055978.
- The clock that counts and the Feynman checkerboard: MM 2.40 and 2.40.1 (DOI 10.5281/zenodo.23280168, 10.5281/zenodo.23406847).
- a_0 and cosmic age: MM 2.42 and 2.42.1 (DOI 10.5281/zenodo.23294652, 10.5281/zenodo.23329094).
- Dark matter and 3√3: MM 2.44, DOI 10.5281/zenodo.23329080.
- The scale $cH_0/6$ and clusters: MM 2.46 and 2.46.1 (DOI 10.5281/zenodo.23340834, 10.5281/zenodo.23506655).
- Born, Tsirelson and the loop: MM 2.48, DOI 10.5281/zenodo.23352814; Born from counting copies: MM 2.48.1, DOI 10.5281/zenodo.23655646.
- Dark energy as a reading of the clock: MM 2.50, DOI 10.5281/zenodo.23376729.
- The Standard Model: MM 2.52, DOI 10.5281/zenodo.23404761.
- The census of constants and the prediction on 3√3: MM 2.54, DOI 10.5281/zenodo.23418156.
- Wide binaries: MM 2.56, DOI 10.5281/zenodo.23506401.
- Primordial deuterium: MM 2.58, DOI 10.5281/zenodo.23521754.
- Cassini, Saturn and the binaries: MM 2.60, DOI 10.5281/zenodo.23533882.
- Objects as the common part of two points of view: MM 2.62, DOI 10.5281/zenodo.23609967.
- The mass threshold and the fractional distance: MM 2.66, DOI 10.5281/zenodo.23643759.
- This overview: DOI 10.5281/zenodo.23535916.

Where to start. This overview, then the entry page 2.27 and volume 2.30. For a single computation to redo: the clock that counts (2.40). For a single prediction with a date: 3√3 (2.54).

Glossary

- **Nexus.** A relation between two “logical differents” which exist for each other thanks to what they have in common.
- **Ends of the nexus.** The Described, the side that never presents itself as an object, and the Descriptor, the side that counts and opens onto the local.
- **GCD.** The element common to the two ends, the only one that makes them exist for each other; in the numerical model, the greatest common divisor, like the 1 between 3 and 5.
- **Non-local phase.** Reality as pure relations, without space, without time and without objects; present everywhere, neither before nor beneath the rest.
- **Local phase.** The same reality read from inside, from a given point: space, time, objects, measurements.
- **Matrix.** The set of repeated copies of one and the same nexus. The multiples, in line, are read as time; the fractions, side by side, as space.
- **Translator.** The passage from counting to measuring, where units, geometry and constants come in.
- **Logical diffraction.** The way the observer's matrix opens out into fractions, in which the object appears.
- **Mirror.** The equals sign that makes a sign exist in what defines it; for the observer, the reflection through which it sees everything else.

- **Common part.** What two records have in common: what each reads in its own record, without looking at the other's, with the same value as the other. For MM it is the object that two observers see together.
- **Decoherence.** The disappearance of the fringes when the environment carries away copies of an object's position; coherence is not destroyed, it passes into the correlations with the environment.
- **Fractional distance.** In the AI's translation, the quantity to be distinguished, measured in the cell with which the reader distinguishes positions.
- **Reading.** An MM interpretation of a measured datum. A reading can fall without the framework falling.
- **Registered.** Written down and fixed with a digital fingerprint and a time before the comparison with the data.

Who did what

The framework and the premises are mine; I am not a professional physicist. The mathematics and the computations were developed with artificial-intelligence assistants (GPT-6 Astra through Codex, Claude and, for some records, Gemini), in a dialogue directed by me. The checks are internal: in the most recent records the rules and predictions were registered before the comparison with the data, and a second assistant verified the main steps. There has been no review by human referees: this overview is written precisely for those who will want to carry it out.

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